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Prompt Engineering Assignment 2

Part 1: Start Simple

I will be using ChatGPT as my LLM for this assignment.

Version 1
Input: Are these reviews positive or negative?
Output:

The reviews contain a mix of positive, negative, and neutral sentiment, rather than being overwhelmingly one or the other.

There are 110 reviews across five products:

- Breeze Wireless Earbuds
- Cozy Knit Throw Blanket
- Hearth Electric Kettle
- Lumen Smart Bulbs (4-Pack)
- Trailmark Hiking Backpack

Some reviews are clearly positive (e.g., praising durability or performance), while others describe defects, poor customer service, or misleading product descriptions.

There are also sarcastic reviews, such as "Fantastic. Just fantastic. Mine lasted a full week before the switch broke." Despite the positive-sounding words, that review is negative.

A few things to keep in mind when classifying them:

1. Positive: The reviewer expresses satisfaction, recommends the product, or praises its features.
2. Negative: The reviewer expresses dissatisfaction, complains about a defect, or indicates they would return the product.
3. Neutral: The review is primarily informational, asks a question, or doesn't express a clear opinion.
4. Mixed: The reviewer likes certain features but dislikes others.

Overall assessment: The dataset has mixed sentiment, with plenty of both positive and negative feedback. Neutral and mixed reviews should be handled separately rather than automatically classified as positive or negative.

I can also break down the sentiment of all 110 reviews by product and provide positive, negative, and neutral counts.

Analysis:

- The model doesn't label each review, just summarizes them.
- The model counted the total number of reviews and the number of products, but it doesn't sort or count any of it.
- The model was still able to recognize some reviews as being neutral or mixed. It even picked up on sarcastic reviews too. When making the bullet points, it notes all of this.

Part 2: Improve the Prompt

Version 2 - Defining the Categories

Input: For each product, categorize each review using the following labels: positive, negative, mixed, informational; where good reviews are labeled positive, bad reviews are labeled negative, mixed reviews are a bit of good and bad, and informational containing info or asking a question.

Output:

I'll categorize all 110 reviews using the four requested labels:

- positive – Overall favorable experience.
- negative – Overall unfavorable experience.
- mixed – Contains both favorable and unfavorable feedback.
- informational – Neutral statements, product details, shipping updates, or questions without a clear positive or negative sentiment.

I'll preserve the original review IDs, product names, and review text, and add a `sentiment` column.

Completed – 110 reviews categorized

Your categorized CSV is ready:

product-reviews-categorized.csv

Original columns preserved + sentiment column added
Download

Categorization summary

Product	Positive	Negative	Mixed	Informational
Breeze Wireless Earbuds	5	7	7	3
Cozy Knit Throw Blanket	8	6	5	3
Hearth Electric Kettle	6	9	5	2
Lumen Smart Bulbs	5	9	5	3
Trailmark Hiking Backpack	8	8	4	2

Total	32	39	26	13

I treated sarcasm as negative when the underlying meaning was clearly a complaint, and neutral factual statements as informational.

Differences: While both the base prompt and this version both had the same amount and type of categories, this version of the prompt gives a modified csv download where all the reviews are labeled as well as a final categorization summary. It is worth noting, however, that some reviews the model deems as “information” are actually “mixed”.

Version 3 - Few-shot Prompting

For this test, I held the categories as a constant (so only prompting for positive and negative). I also had to swap to Claude at this point because I hit the upload cap on the free tier of ChatGPT.

Input: Categorize the following reviews as positive or negative? Here are a few examples:

R011 | Negative
R019 | Negative
R029 | Positive
R042 | Positive

Output:
I categorized all 110 reviews: 62 Positive and 48 Negative. The results are in product-reviews-categorized.csv, which adds `sentiment`, `confidence`, and `note` columns to your original data.

I inferred a few rules from your four examples:

- **Lukewarm counts as Positive.** "Fine for podcasts" (R029) and "Ok for the price" (R042) were labeled Positive, so I treated "fine," "decent," and "good enough" the same way.
- **Delivery and seller problems count as Negative.** A stained item (R011) and the wrong size (R019) were Negative, so I labeled shipping delays, damaged boxes, and similar issues Negative unless the review also said the product was fine.
- **Mixed reviews follow the overall verdict.** For example, "Love the sound, hate the touch controls" (R097) went Negative, while "Great bulbs, terrible app" (R009) went Positive.
- **Sarcasm is read literally.** Reviews like R022 ("Ten stars" about a waterproof pack that soaked the sleeping bag) are Negative. I tagged these ten as `sarcasm`.

Of the 110, 71 were clear-cut. 29 were mixed or lukewarm, and 10 had no real opinion, such as questions ("Does this work with a PS5?") or neutral statements ("They're lightbulbs"). I forced those 10 into a label, so treat them as guesses. They're marked Low confidence in the file, and I'd drop or relabel them if you'd rather not count them.

If you want the rules changed, tell me which. For example, I can put lukewarm reviews in Negative or add a Neutral category, and I'll re-run it.

Differences: For this output, the model is a lot more accurate compared to the previous outputs. This attached csv in the output also included a confidence field in addition to the sentiment field.

Version 4 - Output Format

Input: Categorize the following reviews as positive or negative. Format the results such that the columns are in the following order: review id, product, sentiment, review. Also, sort each entry by product type.

Output:
I'll start by reading the relevant skill, then look at the file.

I categorized all 110 reviews and saved them to product-reviews-categorized.csv. The columns are review_id, product, sentiment, review_text. Rows are sorted by product (A–Z) and then by review ID.

Product	Positive	Negative
Breeze Wireless Earbuds	8	14
Cozy Knit Throw Blanket	11	11
Hearth Electric Kettle	11	11
Lumen Smart Bulbs (4-Pack)	10	12
Trailmark Hiking Backpack	12	10

Because you asked for only positive or negative, I had to force a call on some reviews. I handled them this way:

- **Sarcasm:** I labeled these Negative, for example R022 ("waterproof" pack turned sleeping bag into a sponge), R024, R030, R066, R068, R078, R080, R085, R098 and R108.

- **Mixed reviews:** I went with the overall tone. Praise with a real flaw at the end, like R060, R071, R075 and R097, is Negative. Complaints softened by a clear overall recommendation, like R002, R009, R014 and R020, is Positive.
- **Lukewarm reviews:** "Fine" or "nothing special" is Positive when the product does its job (R007, R042, R043, R092, R104). It is Negative when the reviewer sounds underwhelmed (R016, R029, R038, R073).
- **No opinion on the product:** Questions and unopened gifts (R012, R067, R087, R094, R101) are labeled Negative, since they don't express satisfaction. Shipping-only comments follow their tone, so R026 is Positive and R057, R062 and R093 are Negative.

If you'd rather have a third "Neutral" or "Mixed" label for these, I can re-run it that way.

Differences: The sentiment seems to be the same as the previous prompt, but the output csv has been formatted to organize each review by product type, making it easy to see sentiments by product.

Version 5 - Constraints and Role Prompting
I decided to combine these two just in case I was about to hit the upload cap.

Input: You are someone who analyzes customer feedback and develops reports on both the positive and negative reviews. Analyze these reviews and categorize them, as well as summarize the main positive and negative points. Keep this summary short, about 2 or 3 bullet points.

Output:

Customer Review Analysis: 110 reviews, 5 products

Overall breakdown

Category	Count	Share
Negative	39	35%
Positive	36	33%
Mixed	20	18%

Neutral / no opinion	11	10%
Questions (no opinion)	4	4%

Each product has 22 reviews, and sentiment is fairly even across all five, with about 7 to 9 negative and 7 to 8 positive per product. About 10 negative reviews are sarcastic ("Ten stars," "Truly impressive engineering"), so they read as positive at first glance but are complaints.

Main positives

- **Value and build quality:** Reviewers often say products look and feel pricier than they are, such as the backpack's durability, the blanket's softness and the kettle's fast boil and design.
- **Features people enjoy:** Examples are the earbuds' comfort and battery life, the smart bulbs' scheduling, sunrise alarm and voice control, and the kettle's temperature presets.

Main negatives

- **Early failures and defects:** Items break or die within days to weeks (earbuds disconnecting or failing to charge, kettles stopping or leaking, bulbs going dead, backpack zippers, buckles and stitching, blanket unraveling).
- **Mismatch with listings, plus software and support gaps:** Complaints include the backpack's "waterproof" and size claims, the earbuds' noise cancelling, the bulbs' brightness, and wrong colors or sizes. The Lumen app is a recurring pain point (crashes, Wi-Fi drops, lost settings, location and account requirements), and support is hit-or-miss (kettle and bulb issues were fixed quickly, earbud support never replied).

Smaller signals: Shipping and packaging complaints (damaged or wet boxes, late delivery) and a few safety concerns on the kettle (leaks, hot handle) are worth flagging. The unanswered questions (Google Home compatibility, PS5 support, laptop sleeve size, blanket dimensions) suggest the listings are missing information.

I labeled each review by hand, so the counts are close but judgment calls on mixed and sarcastic reviews could shift them by a few.

Differences: The model didn’t label each entry this time, but it did follow the constraints and instructions laid out by summarizing the main positives and negatives and keeping it to 3 bullet points or less.

Part 3: Check the model's work

For this part, I used the categorizations from Version 2, since that had the most categories. Even if it was more inaccurate than Version 3, it's categories are much more nuanced.

Spotchecking:
R012 | Informational
R015 | Negative
R022 | Negative
R024 | Negative
R026 | Positive
R036 | Mixed
R039 | Positive
R048 | Informational
R098 | Negative
R108 | Negative

Comparing to Version 2's answers:
R012 | Informational
R015 | Negative
R022 | Negative
R024 | Negative
R026 | Informational
R036 | Mixed
R039 | Positive
R048 | Negative
R098 | Negative
R108 | Negative

Analysis:

- The differences between the two were pretty minimal, those being R026 and R048. I think the difference lies in that the model is marking any product malfunction as negative (which is fair) whereas I marked it down as informational unless the reviewer was annoyed in tone. So, to be honest, I kinda agree with the model, since the difference just came down to how I read the tone. I can understand how the model came to its conclusions.
- Changes to the prompt: I definitely think that few-shot prompting would make this more accurate. For this task specifically, adding roles, constraints, or output formatting would change the logic the model uses to judge each review.

- In regards to how each technique affects each output, I think adding more categories and few-shot prompting did the best in providing accurate categorizations. The other 3 affect the summarization of the information from what I saw.
- The model actually handled tricky reviews pretty well, even without specific prompting, which was kinda surprising to me. I was able to pick up sarcastic reviews as negative without having to be provided the examples during the few-shot prompting.
- The model labeled each review except for Versions 1 and 5. I believe this is due to the wording I chose. In Versions 2-4, I specified to categorize each review, and it would output a modified csv file. Whereas with 1 and 5, I didn't specify this, which resulted in the results being summarized instead.
- The technique that made the biggest difference was creating the multiple categories because that meant that the model could handle cases where a review wasn't explicitly positive or negative, such as when the reviewer was asking a question.